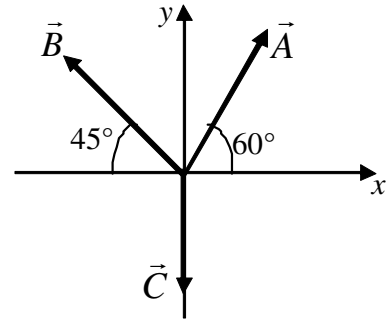


Question 2 /Vraag 2

Consider the vectors in the sketch. Vector $\vec{A}$ has a magnitude of 15 m, vector $\vec{B}$ has a magnitude of 16 m and vector $\vec{C}$ has a magnitude of 12 m as shown in the sketch.

Beskou die vektore in die skets. Vektor A het 'n grootte van 15 m, Vektor B het 'n grootte van 16 m en Vektor C het 'n grootte van 12 m, soos aangedui in die skets.



- a. Write the three vectors using unit vector notation. Skryf die drie vektore in eenheids-vektor notasie.

[3]

$\vec{A} = 15 \cos 60^\circ \hat{i} + 15 \sin 60^\circ \hat{j} = 7.5 \hat{i} + 13 \hat{j} \text{ m}$	✓ components + units
$\vec{B} = 16 \cos 45^\circ \hat{i} + 16 \sin 45^\circ \hat{j} = -11.3 \hat{i} + 11.3 \hat{j} \text{ m}$	✓ components + units
$\vec{C} = 0 \hat{i} - 12 \hat{j} \text{ m}$	✓ C does not have to have $0 \hat{i}$ written there. components + units

- b. Determine the magnitude and direction of the sum of the vectors using the component method.

Bereken die grootte en rigting van die som van die vektore deur gebruik te maak van die component-metode.

[5]

$\vec{A} + \vec{B} + \vec{C} = -3.8 \hat{i} + 12.3 \hat{j} \text{ m}$	✓✓ one each for x and y components – does not have to be unit vector notation
$ \vec{A} + \vec{B} + \vec{C} = \sqrt{(-3.8)^2 + (12.3)^2} = 12.9 \text{ m}$	✓ final value can be rounded to 13 – must have units else 0
Magnitude of resultant vector	
$\tan \theta = \frac{12.3}{-3.8} = -3.24$	✓ one mark for angle and
$\theta = 72.9^\circ$ above the neg. x-axis.	✓ one mark for description above neg x or 163° anti-clockwise from +x axis

Question 4/Vraag 4

Two vectors are given by $\vec{a} = 3.0 \hat{i} + 5.0 \hat{j}$ and $\vec{b} = 2.0 \hat{i} + 4.0 \hat{j}$. Determine the following:

Twee vektore word deur $\vec{a} = 3.0 \hat{i} + 5.0 \hat{j}$ en $\vec{b} = 2.0 \hat{i} + 4.0 \hat{j}$ gegee. Bepaal die volgende:

- (a) $\vec{a} \times \vec{b}$

[3]

$\vec{a} \times \vec{b} = (a_x b_y - a_y b_x) \hat{k}$	✓ for the process of multiplying -
$= [(3.0)(4.0) - (5.0)(2.0)] \hat{k} = 2.0 \hat{k}$	✓✓ one for the final value and one for the direction $\hat{k}$.

- (b) $\vec{a} \cdot \vec{b}$

[3]

$\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y$	✓ for the process of multiplying -
$= (3.0)(2.0) + (5.0)(4.0) = 26.$	✓✓ 2 for the final answer.

Question 5 / Vraag 5

What mass of steam at 100°C must be mixed with 150 g of ice at its melting point, in a thermally insulated container, to produce liquid water at 50 °C.

Hoeveel stoom (massa) by 100 °C word benodig sodat wanneer dit met 150 g ys by die smeltpunt, gemeng word in termies-geïsoleerde houer, dit water met 'n finale temperatuur van 50 °C sal produseer.

[6]

$0 = Q_{ice} + Q_{steam}$ $0 = (mL_F + mc_{water}\Delta T) - ML_v + Mc_{water}\Delta T$ $= m(L_F + c_{water}\Delta T) + M(-L_v + c_{water}\Delta T)$ $-M = \frac{m(L_F + c_{water}\Delta T)}{-L_v + c_{water}\Delta T}$ $M = \frac{0.150 \text{ kg}(333 \text{ J/kg} + 4.187 \text{ J/kg} \cdot \text{K}(+50 \text{ K}))}{2256 \text{ J/kg} - 4.187 \text{ J/kg} \cdot \text{K}(-50 \text{ K})}$ $= 0.033 \text{ kg}$	✓ for the <u>two</u> equations for ice ✓ for the <u>two</u> equations for steam. ✓ ✓ for determining expression for mass of the steam – Not written – no mark ✓ substituting correct values. ✓ final answer in kg or g
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THE END / DIE EINDE

